

ST THOMAS CYRIL KING, Virgin Islands, U.S.

WMO: 785430

Lat: 18.336N

Lon: 64.980W

Elev: 6

StdP: 101.25

Time Zone: -4.00 (W04)

Period: 98-19

WBAN: 11640

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	21.4	22.0	14.2	10.1	24.5	15.7	11.1	24.7	10.2	26.1	9.5	26.1	3.0	80	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	32.1	25.7	31.6	25.6	31.2	25.6	27.2	29.9	26.9	29.8	26.6	29.7	6.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	28.7	25.9	21.3	28.7	25.6	20.9	28.7	85.7	30.3	84.1	29.8	82.8	29.6	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.4	7.9	19.6	33.7	1.9	1.3	18.2	34.7	17.1	35.5	16.0	36.2	14.6	37.2		
			14.6	27.6	3.1	0.6	12.4	28.0	10.6	28.4	8.8	28.8	6.5	29.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	25.6	25.5	25.7	26.5	27.5	28.5	28.8	28.7	28.2	27.2	26.3	(6)	
(7)		DBStd	1.58	0.89	0.80	1.09	0.95	0.99	0.90	0.92	0.97	0.93	1.01	0.94	(7)	
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
(10)		CDD10.0	6302	483	434	486	494	542	555	581	583	562	563	514	505	(10)
(11)		CDD18.3	3261	224	201	227	244	284	305	323	324	312	304	264	247	(11)
(12)		CDH23.3	32991	1552	1331	1657	2181	3001	3603	3857	3935	3758	3463	2625	2028	(12)
(13)		CDH26.7	9584	204	174	241	422	807	1278	1441	1512	1420	1128	611	347	(13)
(14)	Wind	WSAvg	4.5	4.6	4.8	4.6	4.6	4.7	4.9	5.1	4.7	3.9	3.8	4.0	4.7	(14)
(15)	Precipitation	PrecAvg	996	46	36	55	65	98	74	66	100	120	140	118	78	(15)
(16)		PrecMax	1530	94	69	144	417	286	206	143	217	326	430	284	191	(16)
(17)		PrecMin	590	11	13	10	6	9	11	19	20	43	36	20	15	(17)
(18)		PrecStd	275	24	16	39	83	74	56	35	54	59	103	85	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.6	29.6	30.0	30.4	31.3	32.9	33.1	33.2	33.0	31.8	30.9	30.2	(19)
(20)			MCWB	22.6	22.6	23.3	24.2	25.3	25.4	25.2	25.7	26.0	25.5	25.1	24.1	(20)
(21)		2%	DB	28.9	28.9	29.2	29.7	30.6	31.5	31.9	32.0	31.8	31.2	30.3	29.6	(21)
(22)			MCWB	22.6	22.6	23.2	24.1	25.0	25.4	25.4	25.8	26.0	25.5	25.0	24.0	(22)
(23)		5%	DB	28.5	28.5	28.7	29.3	30.1	31.0	31.3	31.4	31.2	30.7	29.9	29.2	(23)
(24)			MCWB	22.7	22.5	23.1	23.9	24.9	25.4	25.4	25.7	25.9	25.5	24.9	23.8	(24)
(25)		10%	DB	28.1	28.0	28.2	28.9	29.6	30.5	30.8	30.9	30.8	30.3	29.5	28.7	(25)
(26)			MCWB	22.7	22.5	22.9	23.8	24.7	25.3	25.3	25.7	25.9	25.5	24.8	23.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.7	24.6	24.8	26.2	26.7	26.9	27.2	27.6	27.6	27.6	26.9	25.7	(27)
(28)			MCDWB	27.3	27.7	27.8	28.9	29.7	30.5	29.9	30.1	30.2	29.8	28.9	28.3	(28)
(29)		2%	WB	24.1	23.8	24.3	25.5	26.1	26.5	26.7	27.0	27.1	27.0	26.2	25.0	(29)
(30)			MCDWB	27.0	27.1	27.4	28.4	29.3	30.2	29.9	30.2	30.0	29.3	28.6	28.0	(30)
(31)		5%	WB	23.7	23.3	24.0	24.9	25.7	26.1	26.4	26.7	26.7	26.5	25.7	24.5	(31)
(32)			MCDWB	26.7	26.8	27.1	27.9	28.9	29.9	29.8	30.0	29.8	29.3	28.3	27.6	(32)
(33)		10%	WB	23.3	23.0	23.5	24.3	25.2	25.7	26.0	26.4	26.4	26.1	25.3	24.1	(33)
(34)			MCDWB	26.5	26.6	26.9	27.6	28.5	29.5	29.6	29.8	29.7	29.3	28.1	27.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	5.1	4.8	4.7	4.4	4.5	5.0	4.9	4.7	4.6	4.8	4.8	(35)
(36)			MCDBR	5.6	5.7	5.3	5.0	4.6	4.8	5.3	5.3	5.0	5.0	5.1	5.3	(36)
(37)			MCWBR	2.4	2.4	2.2	2.2	2.0	2.0	2.1	2.2	2.2	2.1	2.3	2.4	(37)
(38)		5% WB	MCDBR	4.9	5.1	4.6	4.2	4.2	4.5	4.9	4.8	4.6	4.3	4.7	4.7	(38)
(39)			MCWBR	2.5	2.5	2.3	2.2	2.0	2.1	2.2	2.3	2.3	2.3	2.4	2.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.371	0.382	0.383	0.407	0.447	0.515	0.522	0.484	0.440	0.393	0.384	0.376	(40)	
(41)		taud	2.576	2.534	2.530	2.491	2.391	2.159	2.132	2.276	2.428	2.582	2.598	2.594	(41)	
(42)		Ebn at Noon	915	920	927	900	853	790	785	822	860	898	895	898	(42)	
(43)		Edh at Noon	95	103	107	112	122	152	157	137	117	97	92	91	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.76	5.55	6.19	6.55	6.31	6.35	6.44	6.33	5.94	5.21	4.58	4.42	(44)	
(45)		RadStd	0.13	0.12	0.23	0.37	0.42	0.38	0.32	0.23	0.38	0.37	0.32	0.20	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+2.45	+0.32	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.23	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	+79	+80

Nomenclature: See separate page